DB Infrastructure change

Background

As part of capacity planning and to support multiple initiatives, which will introduce load to database in terms of traffic (Read-write), data size, resource utilization, we would need to scale PSP primary database vertically i.e. modify RDS instance size to next or higher instance size and storage. This document covers on planning, test and estimated extra cost.

Configuration

	Config Parameter	Current Value	New Value	Comments
1	Instance Class	db.r5.12xlarge	db.r5.16xlarge	
2	IOPS	40000	45000	
3	Storage - Monolith (us-west-2)	32 TB	37 TB	It depends on allowed % in increase as per RDS
4	ReadReplica - Monolith (us-west-2)	32 TB	37 TB	
5	Storage - Monolith (us-east-2)	32 TB	37 TB	

Benefits

Get more CPU, memory and throughput to serve traffic.

Downtime

Instance size upgrade:

The update will take up to 30 minutes. But the downtime will be about 2-5 minutes in the middle of the change

Adding storage:

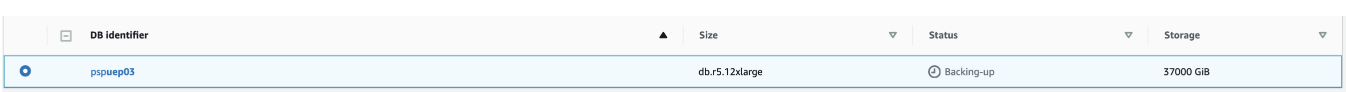
No Downtime in this operation.

Plan

1. On **DR database** for monolith (**pspuep03**): [**CHG1391352**]. We need to test production change and add 5 TB space to monolith DR site database i.e. **pspuep03**
    a. add 5 TB space
    b. Upgrade PSP Monolith DR database instance size to 16xl (from 12xl)
    c. add 5K IOPS. Change IOPS from 40000 to 45000
2. Once change on DR completes without any issue, rollback following change
    a. Rollback PSP Monolith DR database instance size to 12xl (from 16xl)
    b. Rollback IOPS from 45k to 40k
3. Monitor and Ensure no Goldengate replication issue

Output :

- Modification of storage completed in 5 mins. Storage-optimization phase completed in less than 24 hrs.



DB identifier	Size	Status	Storage
pspuep03	db.r5.12xlarge	Backing-up	37000 GiB

- Upgrade PSP Monolith DR database instance size to 16xl (from 12xl)
    - Took **13** mins to complete instance upgrade

	DB identifier	Size	CPU	Current activity	Maintenance	VPC	Multi-AZ
○	pspuep03	db.r5.16xlarge	0.42%	1 Connections	none	vpc-0f237640b6948c5e1	No

- add 5K IOPS. Change IOPS from 40000 to 45000
    - Took 2 mins to for modification.

	DB identifier	Size	Storage	Provisioned IOPS
○	pspuep03	db.r5.16xlarge	37000 GiB	45000

On **Primary production**, Perform this activity over weekend by taking downtime window.

- Take backup of database before change. This will also help in worst case scenario to rollback.
- Enable turn away page to stop incoming traffic
- Modify RDS instance to db.r5.16xlarge
    - The update will take up to 30 minutes. But the downtime will be about 2-5 minutes in the middle of the change
- Once RDS becomes available
- Disable turn away page
- Run basic sanity to confirm all good.
- Add storage by 5 TB (online operation)
- In case Read Replica service goes down. Use below to start:

```
1. Connect to read replica database as below:
intuadmin/"xxxx"@'pspadg01.cjls0bohfgpq.us-west-2.rds.amazonaws.com:1521/pspuwp01'

2. Create service
BEGIN
  DBMS_SERVICE.create_service(
    service_name => 'pspadg01',
    network_name => 'pspadg01'
  );
END;
/

COLUMN name FORMAT A30
COLUMN network_name FORMAT A30
select NAME,NETWORK_NAME,CREATION_DATE,ENABLED from dba_services order by 1;

3. start service
BEGIN
  DBMS_SERVICE.start_service(
    service_name => 'pspadg01'
  );
END;
/

4. Check status of service
SELECT name,       network_name FROM   v$active_services ORDER BY 1;
```

Read Replica:

- Add 5 TB Storage via console
- Monitor and ensure no issue on Read Replica service and replication.

Cost

Instance size upgrade:

There will not be extra cost associated for increasing instance size (to r5.16xlarge) as this would come under bulk RI purchase since this is one instance upgrade.

Adding IOPS & storage:

TBD